

DWSIM Surrogate Model

Results Summary

Benzene–Toluene | Peng–Robinson | DWSIM 9.0.5

0.005244

Mean confirmation NRMSE

0.999493

Mean confirmation R^2

0

Invalid final purities

System and Validation Design

- Dataset: 10,000 converged DWSIM simulations and 340 failed attempts.
- Fixed feed flow: 100 mol/s; thermodynamic model: Peng–Robinson.
- Split: 7,000 training, 1,000 calibration, and 2,000 locked confirmation rows.
- Selected model: 128–64 ReLU ANN with exact benzene-balance projection.
- Selection: lowest five-fold cross-validation mean NRMSE, 0.005410 ± 0.000392 .

Five-Fold Model Comparison

Rank	Model	Mean NRMSE	Mean R^2
1	ANN 128–64 ReLU	0.005410	0.999432
2	ANN 256–128 ReLU	0.005652	0.999376
3	ANN 64–32 ReLU	0.006305	0.999235
4	ANN 128–64 tanh	0.006630	0.999146
5	ANN latent split	0.010421	0.997631
6	CatBoost	0.010601	0.997077
7	Extra Trees	0.021993	0.981505
8	Polynomial Ridge	0.029151	0.980632

Locked Confirmation Metrics

Output	MAE	RMSE	R^2	NRMSE
Distillate benzene purity	0.002004	0.005473	0.999053	0.008259
Bottoms toluene purity	0.001653	0.003570	0.999633	0.005403
Condenser duty (kW)	28.944	37.209	0.999755	0.003330
Signed reboiler duty (kW)	38.852	57.423	0.999529	0.003984

Performance Away from Purity Saturation

Only rows where the corresponding DWSIM purity is below 0.995 are included.

Output	Rows	MAE	RMSE	R^2
Distillate benzene purity	1,133	0.002354	0.006943	0.998419
Bottoms toluene purity	1,084	0.001960	0.004399	0.999380

Sample Predictions: DWSIM versus Surrogate

Row	Source	Distillate purity	Bottoms purity	Condenser kW	Reboiler kW
8980	DWSIM	0.999844	0.629409	3091.34	-3064.56
	Surrogate	0.994596	0.627481	3103.56	-3072.83
7897	DWSIM	0.716766	0.999738	7473.44	-7695.07
	Surrogate	0.717103	1.000000	7441.55	-7663.94
1961	DWSIM	0.999949	0.597290	4392.95	-1031.59
	Surrogate	1.000000	0.597311	4419.14	-1013.52

Physical Consistency

- Invalid final purity predictions: 0.
- Maximum final benzene-balance residual: 1.42×10^{-14} mol/s.
- The raw ANN produced 257 invalid distillate and 250 invalid bottoms predictions before projection.
- Mean projection changes were 0.002250 for distillate purity and 0.001935 for bottoms purity.

Robustness and Direct DWSIM Sweeps

Withheld high-value region	Mean NRMSE
Feed benzene composition	0.004973
Reflux ratio	0.006226
Stage count	0.005190
Bottoms flow	0.011627

At one representative operating point, 13 reflux ratios and all 18 stage counts from 18 to 35 were solved again in DWSIM and predicted independently by the ANN. The duty curves agree closely. Purity differences remain small, but the ANN oscillates near saturation and does not exactly reproduce the DWSIM stage-count plateau. The two comparison figures and operating-point details are in [Report.pdf](#).

Uncertainty and Important Observations

- Nominal interval coverage is 90%; observed coverage is 91.4%, 91.4%, 89.7%, and 89.0% for the four outputs.
- High random-split accuracy is plausible because DWSIM is deterministic, noise-free, and smooth over a space-filling design.
- Simulation failures concentrate at low bottoms flow and high stage count.
- Purity monotonicity is approximate; significant reversals are uncommon but not zero.
- This is an interpolative emulator of the specified DWSIM setup, not a plant model, and it must not be used outside the sampled ranges.